

Mohamed Hossam

backend software engineer

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SUMMARY

Computer Science student focused on backend engineering with Python, Django, and FastAPI. Builds complete systems end to end schema design, service-layer architecture, authentication, and deployment with an emphasis on security and data integrity. Competitive-programming background supports careful reasoning about edge cases, correctness, and performance. Seeking a junior or internship backend role to apply these skills to production systems.

TECHNICAL SKILLS

Backend & APIs: Django, Django REST Framework, FastAPI, RESTful APIs, Celery, Asynchronous Programming, WebSockets, Pydantic

Software Design & Quality: API Design, Service-Layer Architecture, Automated Testing, Performance Profiling, Observability

Databases: PostgreSQL, MySQL, Redis, SQLAlchemy, Schema Design, Indexing & Query Optimization

Infrastructure & Deployment: Docker, Docker Compose, Nginx, Gunicorn, Uvicorn/Daphne, Linux, Git, CI/CD

Languages: Python, SQL, JavaScript, C++, Bash

TECHNICAL PROJECTS

CourseHub — Course Platform

2025 – Present

Django 5.1, PostgreSQL 14, HTMX, Cloudinary, Docker, Nginx

- Designed and built a course platform with session-based access control and a layered architecture (views, service layer, ORM).
- Integrated Cloudinary CDN for media delivery, including signed URLs for private video streaming, adaptive bitrate playback, and automatic image optimization (WebP/AVIF).
- Implemented passwordless email authentication using UUID verification tokens and time-limited OTPs.
- Designed a normalized PostgreSQL schema with indexed lookups for efficient concurrent access.
- Containerized with Docker and served locally through Gunicorn and Nginx.

YouTube Clone

2026 – Present

Django 6, PostgreSQL 16, Redis 7, Celery 5, ClamAV, Docker, ImageKit CDN

- Architected a decoupled 3-layer platform with zero-byte storage on the app server, serving content via signed ImageKit CDN URLs for adaptive HLS (240p–1080p) and on-the-fly watermarked thumbnails; denormalized counters and used `unique_together` for $O(1)$ reads.
- Engineered a defense-in-depth upload pipeline at the trust boundary using python-magic header sniffing, ffprobe container/codec/duration/dimension validation, and SHA-256 deduplication to reject malformed or duplicate files before quarantine.
- Added asynchronous malware scanning with Celery and ClamAV behind a quarantine step, ensuring files remain non-streamable until cleared; implemented retries with backoff for infrastructure failures, with infected files rejected and never exposed to viewers.
- Implemented concurrency-safe abuse controls: Redis-backed token-bucket rate limiting per IP/user via middleware and 24h view deduplication with atomic `F()` increments and a double-check transaction to prevent redundant writes under concurrent load.

FastAPI OCR Microservice

2026 – Present

FastAPI, Tesseract, Docker, Pydantic v2, Prometheus

- Characterized capacity at 7.2 req/s and 890ms p95 under 50 concurrent requests, correlating throughput and tail latency with CPU and queue depth via Prometheus metrics and structured logging.
- Prevented overload by bounding CPU-bound Tesseract behind worker pool and queue backpressure with 429, selecting limits from measured throughput plateau and p99 tradeoffs.
- Hardened stateless FastAPI service for production with 10MB upload limit, 30s timeout, consistent error handling, health/readiness probes, and Docker with automated tests in CI.
- Reduced OCR error rate by 62% (CER 8.4% → 3.1%) via ablation-tested Pillow preprocessing, keeping only stages with proven accuracy gain vs latency cost.

OPEN SOURCE CONTRIBUTIONS

NetBox — Merged PR #22834

Aug 2026

- Enforced context-aware sanitization in `clean_html()`: restricted `img[src]` to HTTP(S)/relative URLs via `nh3 attribute_filter + regex` while `a[href]` retains full `ALLOWED_URL_SCHEMES` — closed `ssh:/file:/ftp:` bypass across 3 rendering paths (Markdown, CustomLink, RSS).
- Shipped 7 regression tests covering whitespace/control-char evasion, javascript: blocking, and scheme-intersection enforcement.

Django CRM — Merged PRs #499 · #456

Jun–Aug 2026

- Added `Product.main_image` end-to-end: new `ImageField` with migration `product_main_image`, lifecycle cleanup on delete/replace, auto-resize to 200×200 PNG with old-file deletion, and admin preview/thumbnail in changelist — production-safe file handling with no orphaned media.
- Fixed `TransactionQualityEvent` admin UX: hid attachment icon when no files exist and added stacked inline template + `file_links_for` template tag to render XSS-safe file links on the Deal page — eliminated false-positive UI state.

LEADERSHIP & COMMUNITY

Leader, Mentor & Problem Setter — ICPC Nahda University Community

2025 – Present

- Mentored 600+ competitive programmers, helping them turn data-structures and algorithms knowledge into practical problem-solving instincts.
- Authored original problems for internal training contests and practice sheets, aligned with ICPC difficulty standards.
- Played a key role in building a strong technical culture within the university's competitive programming community.

ACHIEVEMENTS & CERTIFICATIONS

- GitHub: Ranked #7 in Egypt by GitHub activity, according to Committers.top.
- QECPC 2025: Honorable Mention • ECPC Finalist 2026
- Constructor Open Cup 2026: Ranked 20th internationally (Constructor University & JetBrains Foundation)
- Codeforces ([profile](#)) Candidate Master rating (top 45 in Egypt); solved 2,400+ algorithmic problems across Codeforces, LeetCode, and vJudge
- Certifications: ALX Professional Foundations; ALX Ventures Freelance Academy

EDUCATION

B.Sc. in Computer Science — Nahda University, Beni Suef

Expected 2028

GPA: 3.26 / 4.0